

MoleCheck

On-Device AI Skin-Lesion Screening

Capstone Project Report

Classifying moles as benign or malignant from a phone photo, with a model that runs entirely on the device.

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1. Executive Summary

MoleCheck is a mobile app that helps someone judge whether a mole looks benign or potentially malignant. The user photographs one skin lesion, and a convolutional neural network estimates how likely it is to be malignant. The model runs **entirely on the phone**: the photo is never uploaded, which protects privacy and lets the app work offline.

On a held-out test set of 1,722 images the classifier reaches a **ROC-AUC of 0.914**. Tuned for screening, it catches **90% of malignant lesions** while correctly clearing about **75% of benign** ones. The model was exported to TensorFlow Lite (verified faithful to within 0.001 probability) and embedded in a Flutter app tested end-to-end on a device.

MoleCheck is an educational project, not a medical device, and is designed throughout to send users to a dermatologist rather than replace one.

2. Introduction & Motivation

Skin cancer is among the most treatable cancers when caught early and most dangerous when caught late, yet access to dermatology is uneven and most people cannot tell which moles are worth showing a doctor. A phone triage aid that flags concerning lesions could lower the barrier to seeking care.

The goal was to build such an aid end-to-end: train an image classifier on a public dermatology dataset, package it to run on a phone with no server, and wrap it in an app a non-expert can use. The project spans machine learning, model engineering (export and quantisation), and mobile development.

3. Data

Training used the HAM10000 collection from the ISIC Archive — public dermoscopic images of pigmented lesions (11,720 images, each labelled Benign, Malignant, or Indeterminate). The task was framed as binary classification; indeterminate images were dropped, leaving a realistically imbalanced set (~18% malignant).

Preventing data leakage

The same physical lesion is often photographed several times. Splitting those duplicates naively across train and test would let the model memorise specific lesions and post inflated scores, so the data was split **by lesion, not by image** — every photo of a lesion stays in one split.

Split	Lesions	Images	Malignant %
Train	6,121	8,123	18.8%
Validation	1,309	1,726	18.1%
Test	1,309	1,722	18.2%

4. Methodology

The classifier is a **YOLO11s-cls** convolutional network (5.4M parameters), pretrained on ImageNet and fine-tuned on the mole data. It was chosen because it transfer-learns quickly on a modest GPU and exports cleanly to mobile formats. Training ran 40 epochs on an NVIDIA RTX 5060 Ti in about six minutes, with heavy flip/rotation/colour augmentation to combat the class imbalance and mimic the variability of real phone photos.

5. Results

Plain accuracy is misleading here — always guessing 'benign' already scores ~82%. The meaningful metrics are sensitivity, specificity, and the threshold-independent ROC-AUC.

Metric	Value	Meaning
ROC-AUC	0.914	Ranks malignant above benign well
Sensitivity	90.1%	Malignant lesions correctly flagged
Specificity	74.7%	Benign lesions correctly cleared

Choosing the decision threshold

Missing a cancer is far worse than a false alarm, so the default 0.5 cutoff is wrong for screening. The threshold was lowered to **0.137** to reach ~90% sensitivity. The app deliberately errs toward caution.

6. Deployment

Running on-device required converting PyTorch to TensorFlow Lite. Because the direct export is unsupported on Windows, the model was routed PyTorch → ONNX → onnx2tf → TFLite. Correctness was **verified, not assumed**: across 32 test images the exported model matched the original to within 0.0009 probability with identical decisions. The 20.8 MB model ships inside a Flutter app (one codebase for iOS and Android) with a three-screen disclaimer onboarding, and was tested end-to-end on an Android emulator.

7. Limitations & Ethical Considerations

- Not a medical device — an educational prototype, not validated for clinical use.
- It misses some cancers: at 90% sensitivity, ~1 in 10 malignant lesions is not flagged.
- Domain shift: trained on dermatoscope images; everyday phone photos are harder.
- Public datasets under-represent darker skin tones — a known equity concern.

These limitations shape the design: the app is framed as a prompt to seek professional care, never a substitute for it.

8. Future Work

- Fine-tune on ordinary smartphone photos to close the domain gap.

- Measure and mitigate performance across skin tones.
- Add calibrated uncertainty so the app can say 'image too low-quality to assess'.
- Ship the iOS build and run a small usability study.